

Experiment harness
creates 12 annotated
gated training Jobs

1. submit manifests

Kubernetes API
stores desired and
observed cluster state

2. watch burst

KubeML-Aware
watches complete burst
validates, ranks, removes gate

3. remove gate

default-scheduler
checks feasibility
and assigns a node

4. bind

Kubelet
asks containerd
to start each Pod

5. launch

Training Pods
NumPy compute, buffers,
copies, checkpoint I/O

6. events

Evidence store
release, start, completion,
rank and resource samples

fresh CPU evidence

Metrics API
fresh node CPU
and allocatable capacity

Kubernetes provides the API, placement engine, and node runtime.

KubeML-Aware changes **when** a training Pod becomes eligible; it does not replace placement or container execution.